

PATTERSON, LA, USA

WMO: 722329

Lat: 29.717N

Lon: 91.333W

Elev: 9

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.9	(c) 35.5	(d) 6.8	(e) 7.8	(f) 55.4	(g) 12.0	(h) 10.1	(i) 59.9	(j) 22.7	(k) 61.8	(l) 20.4	(m) 61.2	(n) 6.2	(o) 20	(p) 0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	93.0	76.1	91.2	76.8	90.3	76.7	81.4	88.8	80.5	87.7	79.7	86.9	7.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	152.4	85.1	78.8	149.4	84.5	77.7	144.1	83.5	44.9	88.9	44.0	87.8	43.1	86.8	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years				
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	17.2	15.6	26.4	95.5	5.5	3.6	22.5	98.1	19.3	100.2	16.2	102.2	12.3	104.9
			22.7	82.4	4.5	2.2	19.4	84.0	16.8	85.3	14.3	86.5	11.0	88.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	53.7	57.5	62.9	68.5	75.4	80.4	82.2	82.1	79.2	70.3	61.0	55.6	(6)
	DBStd	12.27	9.81	9.31	7.94	6.18	4.65	2.54	2.42	2.35	4.02	6.89	8.06	9.19	(7)
	HDD50	171	75	32	9	0	0	0	0	0	0	0	9	46	(8)
	HDD65	1299	367	240	133	38	3	0	0	0	0	32	169	317	(9)
	CDD50	7154	189	242	409	556	789	913	998	995	876	628	340	219	(10)
	CDD65	2805	17	29	68	143	326	463	533	530	426	196	50	24	(11)
	CDH74	24091	24	69	241	840	2650	4432	5390	5153	3573	1410	239	70	(12)
	CDH80	8702	0	2	7	96	806	1780	2220	2066	1351	355	18	1	(13)
(14) Wind	WSAvg	6.6	8.2	8.3	8.0	7.9	6.9	5.7	4.8	4.5	5.4	6.0	6.6	7.4	(14)
(15) Precipitation	PrecAvg	59.8	5.2	3.5	3.7	3.6	4.3	6.8	7.0	6.8	5.2	4.7	4.2	4.7	(15)
	PrecMax	89.9	17.4	9.2	11.1	9.9	17.3	21.9	9.9	13.3	13.9	12.5	14.3	14.8	(16)
	PrecMin	46.5	0.4	0.6	0.1	0.1	0.2	1.6	3.9	3.0	1.6	0.4	0.7	1.9	(17)
	PrecStd	11.5	4.3	2.6	2.5	2.8	3.7	4.3	1.6	2.7	2.5	3.3	3.4	3.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	79.3	81.2	84.3	90.7	93.3	97.1	95.4	95.4	88.5	82.1	79.1	(19)
		MCWB	68.7	70.1	69.9	70.9	68.7	73.4	67.1	80.0	79.1	75.4	71.2	70.8	(20)
	2%	DB	73.3	75.7	79.2	82.1	88.4	91.0	92.9	92.6	91.1	86.2	79.5	75.4	(21)
		MCWB	66.8	68.5	69.0	71.1	69.3	74.8	74.1	79.0	78.2	73.9	69.6	68.6	(22)
	5%	DB	72.0	73.2	76.9	81.2	86.3	90.0	91.0	90.8	89.8	83.8	77.0	73.0	(23)
		MCWB	66.3	67.0	68.2	70.5	72.1	75.8	76.3	78.5	77.6	72.4	68.0	66.6	(24)
	10%	DB	69.6	71.6	74.6	79.3	84.1	88.3	90.0	89.8	87.8	81.6	74.1	70.4	(25)
		MCWB	64.8	66.1	67.0	68.8	72.2	75.9	76.5	78.4	76.3	70.1	66.5	65.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.1	72.9	73.2	75.7	78.4	81.7	82.4	82.5	81.8	79.1	75.0	73.7	(27)
		MCDB	74.7	77.0	78.6	80.9	85.6	88.3	90.8	91.4	89.6	84.5	79.3	77.2	(28)
	2%	WB	69.3	70.9	71.4	74.4	77.4	80.3	81.2	81.4	80.3	77.1	73.0	71.0	(29)
		MCDB	72.6	74.2	75.7	79.9	84.1	87.0	88.7	89.0	87.5	83.1	77.1	73.8	(30)
	5%	WB	67.1	69.1	70.2	73.2	76.4	79.3	80.2	80.4	79.2	75.4	70.8	69.0	(31)
		MCDB	70.5	72.2	74.6	78.7	82.9	86.0	87.6	87.8	86.2	80.9	75.1	72.5	(32)
	10%	WB	64.8	67.0	68.7	71.8	75.3	78.3	79.4	79.5	78.1	73.7	68.6	66.3	(33)
		MCDB	67.9	70.2	73.4	76.8	81.6	85.3	86.6	86.7	84.9	78.5	73.1	69.5	(34)
(35) Mean Daily Temperature Range	MDBR		17.7	17.1	17.6	17.7	16.8	15.5	14.9	15.2	15.8	19.0	19.6	18.1	(35)
	5% DB	MCDBR	17.0	16.2	16.5	17.1	17.6	17.3	17.0	16.7	17.5	18.9	19.2	18.5	(36)
		MCWBR	12.2	10.7	9.5	9.2	8.2	7.6	8.0	7.5	7.5	9.5	12.1	13.4	(37)
	5% WB	MCDBR	15.0	14.3	14.5	14.5	14.4	14.5	15.1	15.4	15.3	15.4	16.5	16.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.343	0.358	0.378	0.423	0.449	0.467	0.496	0.491	0.441	0.385	0.360	0.352	(40)
	taud		2.515	2.481	2.412	2.290	2.251	2.249	2.199	2.212	2.319	2.460	2.471	2.498	(41)
	Ebn at Noon		283	288	288	277	269	263	255	255	265	275	274	273	(42)
	Edh at Noon		28	32	36	43	44	44	46	45	39	32	29	28	(43)
(44) All-Sky Solar Radiation	RadAvg		952	1126	1525	1845	2028	1967	1837	1734	1628	1453	1099	854	(44)
	RadStd		89	150	136	134	107	153	127	150	140	156	104	80	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (18 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	+230	+155	(47)

Nomenclature: See separate page